Edge-Geometric-Arithmetic and Edge-Mostar Indices of Carbon Nanocones

Guangfu Wang* and Ruiqian Cui

School of Mathematics and Information Sciences, Yantai University,
Yantai, Shandong Province, 264005, China

Abstract

The Mostar index and its edge variant quantify distance nonbalancedness of bonds in a molecular graph. In contrast, geometric-arithmetic type descriptors measure a normalized balance by comparing geometric and arithmetic means of local structural quantities. Motivated by the second geometric-arithmetic index and by the edge-Mostar index, we study the edge-distance geometric-arithmetic index

$$\mathrm{GA}_e(G) = \sum_{e=uv\in E(G)} \frac{2\sqrt{m_u(e)m_v(e)}}{m_u(e)+m_v(e)},$$

where $m_u(e)$ and $m_v(e)$ are the numbers of edges closer to u and v, respectively. We prove a quotient formula for this descriptor over strength-weighted quotient graphs and apply it to the carbon nanocone family $\mathrm{CNC}_m(n)$, whose core is an m-cycle surrounded by n layers of hexagons. An exact summation formula for $\mathrm{GA}_e(\mathrm{CNC}_m(n))$ is obtained, together with an exact absolute-value expression for $\mathrm{Mo}_e(\mathrm{CNC}_m(n))$. For the sign-stable range, in particular for $m \geq 6$, the usual polynomial formula for the edge-Mostar index is recovered. Numerical values for $\mathrm{CNC}_5(n)$, $\mathrm{CNC}_6(n)$, and $\mathrm{CNC}_7(n)$ are reported and compared. The results show that GA_e records normalized edge-distance balance, whereas Mo_e records unnormalized edge-distance nonbalancedness and grows much faster with the number of hexagonal layers.

Keywords: carbon nanocone; edge-geometric-arithmetic index; edge-Mostar index; molecular graph; quotient graph; topological index.
MSC 2020: 05C09; 05C12; 05C92.

1 Introduction

Topological indices translate a molecular graph into numerical descriptors that can be used in mathematical chemistry, quantitative structure-property studies, and structural comparison of chemical networks. In a chemical graph, vertices represent atoms and edges represent bonds. Classical bond-additive descriptors, including Wiener-type, Szeged-type, Padmakar-Ivan-type, and Mostar-type indices, are particularly useful because their local contributions can often be computed by cut methods.

The geometric-arithmetic index was introduced by Vukicevic and Furtula as a degree-based descriptor formed from the ratio between the geometric and arithmetic means of endpoint degrees [2]. A distance-based version, usually called the second geometric-arithmetic index, was proposed by Fath-Tabar, Furtula, and Gutman [3]; it replaces degrees by the quantities $n_u(e)$ and $n_v(e)$, the numbers of vertices closer to the two endpoints of an edge, and has the form

$$GA_2(G) = \sum_{e=uv\in E(G)} \frac{2\sqrt{n_u(e)n_v(e)}}{n_u(e)+n_v(e)}.$$

*Corresponding author. E-mail: gfwang@ytu.edu.cn.

This viewpoint was used, for example, to compare geometric-arithmetic and Mostar indices for graphs obtained from path graph operations [6]. In this paper, the phrase edge-geometric-arithmetic index is used in an edge-distance sense, because the local data are $m_u(e)$ and $m_v(e)$. This should be distinguished from the degree-based edge GA index defined through line-graph degrees [9].

The Mostar index, introduced by Doslic et al. [4], is defined by summing $|n_u(e)-n_v(e)|$ over all edges $e = uv$. It measures distance nonbalancedness of bonds. Arockiaraj, Clement, and Tratnik later introduced the edge-Mostar and total-Mostar variants and developed a strength-weighted quotient-graph technique for carbon nanocones and coronoid systems [5]. The edge-Mostar index is obtained by replacing vertex counts $n_u(e), n_v(e)$ by edge counts $m_u(e), m_v(e)$.

The purpose of the present article is to combine these two directions. We define and compute an edge-geometric-arithmetic descriptor based on $m_u(e)$ and $m_v(e)$, and we compare it with the edge-Mostar index. The chosen graph family is the carbon nanocone $\mathrm{CNC}_m(n)$. Carbon nanocones are chemically meaningful graphene-based nanostructures obtained by removing a wedge from a graphite sheet and joining the boundary rays; their mathematical models consist of an m-cycle core surrounded by n layers of hexagons [10, 5]. This gives a natural setting in which edge-distance balance can be studied as a function of the cone parameter m and the number of layers n. The visual material is used only for orientation and notation: Figure 1 explains the geometric formation of a carbon nanocone, and Figure 2 gives a layer-by-layer counting diagram for the peripheral class $E_{j,i}$. The enumerations below are based on this c-partition and the strength-weighted quotient graph in Lemma 4.1, not on a metric drawing of a three-dimensional cone.

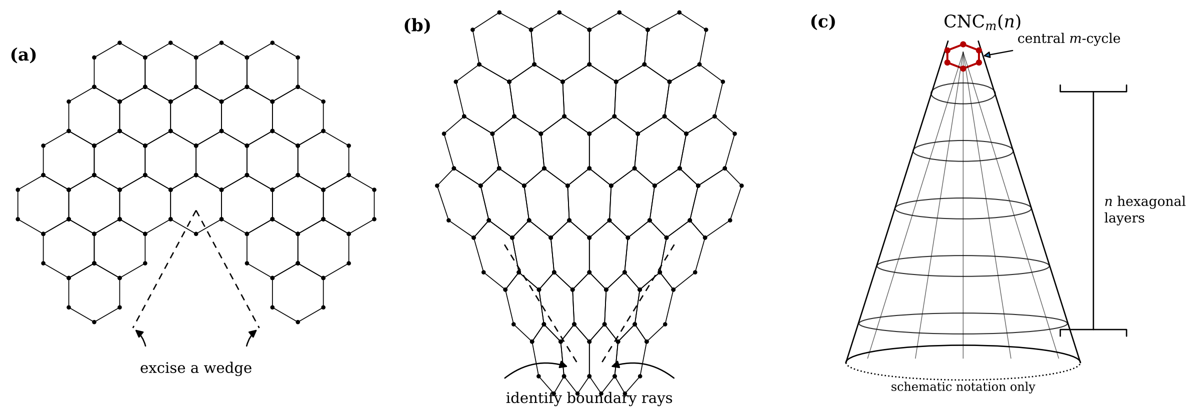

Figure 1: Schematic formation of the finite carbon nanocone graph $\mathrm{CNC}_m(n)$. Panel (a) shows the excision of a wedge from a graphene-like hexagonal sheet; panel (b) indicates the identification of the two boundary rays; panel (c) records only the notation of the finite conical model. The central cycle of length m and the n surrounding hexagonal layers are labelled, while exact edge counts are derived from the quotient diagram in Figure 2.

The main contributions are as follows. First, we define GA_e and prove its quotient formula for strength-weighted graphs. Second, we obtain an exact formula for $\mathrm{GA}_e(\mathrm{CNC}_m(n))$. Third, we give an exact absolute-value expression for $\mathrm{Mo}_e(\mathrm{CNC}_m(n))$ and state the polynomial form obtained in the sign-stable range. Finally, we provide numerical comparisons for the pentagonal, hexagonal, and heptagonal nanocone families.

2 Preliminaries

Throughout the paper, all graphs are finite, simple, and connected. For a graph G, let $V(G)$ and $E(G)$ be its vertex and edge sets. The usual shortest-path distance between vertices $x, y \in V(G)$

is denoted by $d_G(x, y)$. If $f = ab \in E(G)$ is an edge and $x \in V(G)$, then

$$d_G(x, f) = \min\{d_G(x, a), d_G(x, b)\}.$$

For an edge $e = uv \in E(G)$, define

$$\begin{aligned} N_u(e) &= \{x \in V(G) : d_G(u, x) < d_G(v, x)\}, \\ N_v(e) &= \{x \in V(G) : d_G(v, x) < d_G(u, x)\}, \\ M_u(e) &= \{f \in E(G) : d_G(u, f) < d_G(v, f)\}, \\ M_v(e) &= \{f \in E(G) : d_G(v, f) < d_G(u, f)\}. \end{aligned}$$

Let

$$n_u(e) = |N_u(e)|, \qquad n_v(e) = |N_v(e)|,$$

and

$$m_u(e) = |M_u(e)|, \qquad m_v(e) = |M_v(e)|.$$

The edge-Mostar index is

$$\mathrm{Mo}_e(G) = \sum_{e=uv \in E(G)} |m_u(e) - m_v(e)|. \tag{1}$$

2.1 The edge-geometric-arithmetic index

We define the edge-geometric-arithmetic index of G by

$$\mathrm{GA}_e(G) = \sum_{e=uv \in E(G)} \frac{2\sqrt{m_u(e)m_v(e)}}{m_u(e) + m_v(e)}, \tag{2}$$

whenever $m_u(e) + m_v(e) > 0$ for all edges under consideration. The carbon nanocones studied below satisfy this condition. If one wants a convention for exceptional isolated-bond cases, the corresponding undefined term may be omitted or set to 0; this convention is immaterial for the present family.

For each edge $e = uv$, put

$$D(e) = |m_u(e) - m_v(e)|, \qquad S(e) = m_u(e) + m_v(e).$$

Then the elementary identity

$$\frac{2\sqrt{m_u(e)m_v(e)}}{m_u(e) + m_v(e)} = \sqrt{1 - \left(\frac{D(e)}{S(e)}\right)^2} \tag{3}$$

holds for $S(e) > 0$. Therefore Mo_e and GA_e measure the same edge-distance imbalance from complementary perspectives: Mo_e sums the unnormalized difference $D(e)$, whereas GA_e sums a decreasing function of the normalized difference $D(e)/S(e)$.

Proposition 2.1. *If $m_u(e) + m_v(e) > 0$ for every edge of G, then*

$$0 \le \mathrm{GA}_e(G) \le |E(G)|.$$

Moreover, $\mathrm{GA}_e(G) = |E(G)|$ *if and only if* $m_u(e) = m_v(e)$ *for every edge* $e \in E(G)$.

Proof. For positive $x + y$, the inequality $2\sqrt{xy} \le x + y$ gives each summand in (2) at most 1 and at least 0. Equality in every summand occurs exactly when $x = y$, that is, when $m_u(e) = m_v(e)$ for every edge e. □

3 A quotient formula for GA_e

We use the strength-weighted quotient setting of Arockiaraj, Clement, and Tratnik [5]. For the edge-distance quantities considered here, it is enough to keep track of vertex strengths and edge strengths. We write

$$G_{sw} = (G, s_v, s_e),$$

where s_v is a nonnegative vertex strength and s_e is a nonnegative edge strength. In the ordinary unweighted molecular-graph case, $s_e \equiv 1$ and $s_v \equiv 0$, so the side strength below is just the number of nearer edges.

For a strength-weighted graph and an edge $e = uv$, define

$$m_u(e \mid G_{sw}) = \sum_{x \in N_u(e)} s_v(x) + \sum_{f \in M_u(e)} s_e(f), \tag{4}$$

and analogously $m_v(e \mid G_{sw})$. Thus, in the ordinary unweighted case, (4) reduces to the number of edges closer to u than to v.

For a strength-weighted graph, define

$$\mathrm{GA}_e(G_{sw}) = \sum_{e=uv \in E(G)} s_e(e) \frac{2\sqrt{m_u(e \mid G_{sw}) m_v(e \mid G_{sw})}}{m_u(e \mid G_{sw}) + m_v(e \mid G_{sw})}. \tag{5}$$

For two edges $e = uv$ and $f = xy$ of a connected graph G, the Djoković-Winkler relation is defined by

$$e \,\Theta\, f \quad \Longleftrightarrow \quad d_G(u,x) + d_G(v,y) \neq d_G(u,y) + d_G(v,x).$$

The relation Θ is reflexive and symmetric, but it need not be transitive. Its transitive closure, denoted by Θ^*, is the smallest transitive relation on $E(G)$ that contains Θ. A partition $\mathcal{E} = \{E_1, \ldots, E_k\}$ of $E(G)$ is called a *c*-partition if every E_i is a union of Θ^*-classes. For $E_i \in \mathcal{E}$, the quotient graph G/E_i has as vertices the connected components of $G - E_i$; two components X and Y are adjacent in G/E_i whenever some edge of E_i has one endpoint in X and the other in Y.

We shall use the following quotient-weight convention. If X is a component of $G - E_i$, put

$$s_v^i(X) = \sum_{x \in V(X)} s_v(x) + \sum_{f \in E(X)} s_e(f), \tag{6}$$

where $E(X)$ is the set of original edges with both endpoints in X. If XY is an edge of G/E_i, define

$$s_e^i(XY) = \sum_{f \in E_i,\, f \text{ joins } X \text{ and } Y} s_e(f). \tag{7}$$

The resulting strength-weighted quotient graph is denoted by G_{sw}/E_i.

Lemma 3.1 (weighted quotient side equality)**.** *Let $\mathcal{E}$ be a c-partition of $E(G)$, fix $E_i \in \mathcal{E}$, and let $e = uv \in E_i$. Let U and V be the components of $G - E_i$ containing u and v, respectively, and let $E = UV$ be the corresponding quotient edge in G/E_i. Then*

$$m_u(e \mid G_{sw}) = m_U(E \mid G_{sw}/E_i), \qquad m_v(e \mid G_{sw}) = m_V(E \mid G_{sw}/E_i). \tag{8}$$

In particular, all original edges represented by the same quotient edge have the same two side weights, up to interchanging the two sides.

Proof. We spell out the "three-side" agreement used in the quotient method. For the quotient edge $E = UV$, every component X of $G - E_i$ lies in exactly one of the following three classes: closer to U than to V, closer to V than to U, or equidistant from U and V in the quotient graph. Because E_i is a union of Θ^*-classes, this trichotomy agrees with the original distances to u and v: all vertices of such a component X are respectively closer to u, closer to v, or equidistant from u and v. The same assertion holds for the internal edges of X, since the distance from u or v to an internal edge is the minimum of the distances to its endpoints. Edges joining two different components of $G - E_i$ are precisely represented by quotient edges and are counted through the quotient edge strength in (7).

Therefore the total s_v-weight of vertices and the total s_e-weight of edges on the u-side of e aggregate exactly to the side weight on the U-side of E in G_{sw}/E_i, with s_v^i and s_e^i defined by (6) and (7). This gives the first equality in (8); the second follows by symmetry. □

Theorem 3.2. *Let G_{sw} be a strength-weighted graph and let $\mathcal{E} = \{E_1, \dots, E_k\}$ be a c-partition of $E(G)$. Then*

$$\mathrm{GA}_e(G_{sw}) = \sum_{i=1}^{k} \mathrm{GA}_e(G_{sw}/E_i). \tag{9}$$

Proof. Fix i and a quotient edge $E = UV$ of G_{sw}/E_i. Let $\widehat{E}$ be the set of original edges of E_i represented by E. By Lemma 3.1, each $e \in \widehat{E}$ has side weights equal to $m_U(E \mid G_{sw}/E_i)$ and $m_V(E \mid G_{sw}/E_i)$, possibly in the reverse order. Since the geometric-arithmetic factor is symmetric in its two arguments,

$$\sum_{e \in \widehat{E}} s_e(e) \frac{2\sqrt{m_u(e)m_v(e)}}{m_u(e) + m_v(e)} = s_e^i(E) \frac{2\sqrt{m_U(E)m_V(E)}}{m_U(E) + m_V(E)}.$$

Summing this equality over all quotient edges of G/E_i and then over $i = 1, \dots, k$ proves (9). □

4 Carbon nanocones

The carbon nanocone $\mathrm{CNC}_m(n)$ has an m-cycle as its core and is surrounded by n layers of hexagons. We assume throughout this section that m, n are integers with $m \geq 5$ and $n \geq 1$.

The order and size of $\mathrm{CNC}_m(n)$ are

$$|V(\mathrm{CNC}_m(n))| = m(n+1)^2, \qquad |E(\mathrm{CNC}_m(n))| = \frac{m(n+1)(3n+2)}{2}. \tag{10}$$

Here is a direct counting explanation for (10). The core contributes m vertices and m edges. When the kth hexagonal layer is attached, $1 \leq k \leq n$, each of the m sectors contributes $2k+1$ new vertices and $3k+1$ new edges. Hence

$$|V| = m + m\sum_{k=1}^{n}(2k+1) = m(n+1)^2,$$

while

$$|E| = m + m\sum_{k=1}^{n}(3k+1) = \frac{m(n+1)(3n+2)}{2}.$$

The redrawn counting diagram in Figure 2 fixes the notation used for the peripheral quotient classes and, in particular, makes the identity $r_i = n + 1 + i$ visually checkable.

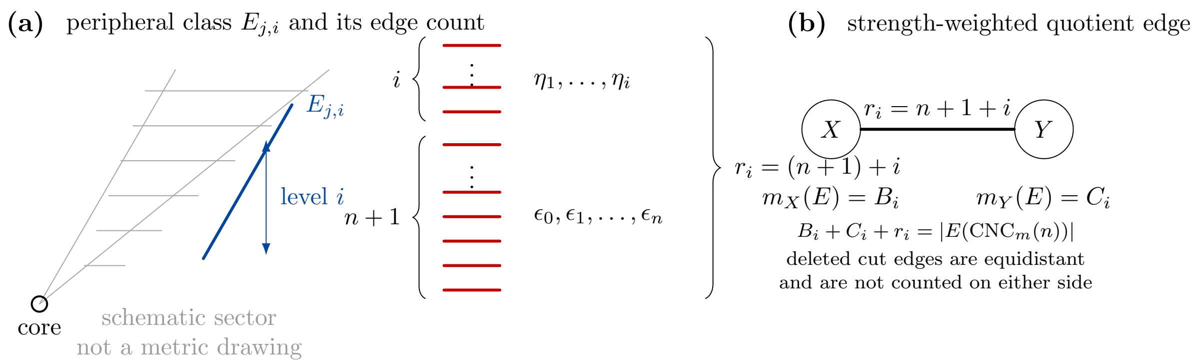

Figure 2: Quotient-class and edge-counting diagram for $\mathrm{CNC}_m(n)$. Panel (a) is a counting diagram for a typical peripheral Θ^*-class $E_{j,i}$: the class contains $n+1$ boundary-side bonds $\epsilon_0, \ldots, \epsilon_n$ and i additional layer bonds $\eta_1, \ldots, \eta_i$, so its quotient edge strength is $r_i = |E_{j,i}| = n+1+i$. Panel (b) shows the resulting strength-weighted quotient edge used in Lemma 4.1; the deleted cut edges are equidistant from the two quotient endpoints and are therefore not included in either side weight B_i or C_i.

We call a Θ^*-class *peripheral* if it is one of the cut classes running from the core toward the boundary of the nanocone. Such classes are denoted by $E_{j,i}$, where $j \in \{1, \ldots, m\}$ indicates the rotational direction and $i \in \{1, \ldots, n\}$ indicates the layer level. The remaining Θ^*-classes are called *core classes* because they are supported on the central cycle.

Lemma 4.1 (quotient data for $\mathrm{CNC}_m(n)$)**.** *In the standard c-partition of $\mathrm{CNC}_m(n)$, for each level $i = 1, \ldots, n$ there are m symmetric peripheral classes. The quotient graph of a peripheral class $E_{j,i}$ is a single edge of strength*

$$r_i = n + 1 + i. \tag{11}$$

For the two components obtained after deleting this class, the two edge-side strengths are

$$B_i = \frac{3i^2 + (6n+3)i - 2n - 2}{2}, \tag{12}$$

and

$$C_i = \frac{m(n+1)(3n+2) - 3i^2 - (6n+5)i}{2}. \tag{13}$$

The core classes are balanced with respect to the two edge sides, and their total edge strength is $m(n+1)$.

Proof. Fix a direction j and a level i. Deleting $E_{j,i}$ separates the nanocone into two components; in the quotient this is represented by one edge XY. By the quotient-weight convention (7), the strength of XY is the total edge strength of all original edges in $E_{j,i}$. Since the present nanocone graph is unweighted, each original bond has strength 1, and hence

$$r_i = s_e^{j,i}(XY) = \sum_{e \in E_{j,i}} 1 = |E_{j,i}|.$$

It remains to count the bonds of this peripheral Θ^*-class. As shown in Figure 2(a), the class $E_{j,i}$ consists of two disjoint groups:

$$E_{j,i} = \{\epsilon_0, \epsilon_1, \ldots, \epsilon_n\} \,\dot{\cup}\, \{\eta_1, \ldots, \eta_i\}.$$

The first group contains one bond in each of the $n+1$ boundary-side positions of the finite cone. The second group contains one additional bond for each of the first i layer steps from the initial peripheral class to level i. Therefore

$$|E_{j,i}| = (n+1) + i,$$

which gives the quotient edge strength

$$r_i = n + 1 + i.$$

Equivalently, $r_1 = n+2$ and moving from level i to level $i+1$ adds exactly one bond to the same peripheral class, so $r_{i+1} = r_i + 1$ and hence $r_i = n+1+i$.

We next compute the nearer-side edge strength. The component on the B_i-side can be counted strip by strip. The ℓth strip, $1 \le \ell \le i$, contributes $3(n+\ell)$ edge-strength units, and the common boundary contribution $n+1$ is not counted on either side because the deleted class is equidistant from the two quotient endpoints. Thus

$$B_i = \sum_{\ell=1}^{i} 3(n+\ell) - (n+1) = \frac{3i^2 + (6n+3)i - 2n - 2}{2}.$$

This gives (12). Every original edge is now either on the B_i-side, on the C_i-side, or in the deleted cut class of strength r_i. Therefore

$$B_i + C_i + r_i = |E(\mathrm{CNC}_m(n))|.$$

Substituting (10), (11), and (12) into this identity yields (13).

Finally, the core classes are invariant under the rotational symmetry of the central m-cycle. Their quotient graphs have equal side strengths, so they contribute no edge-Mostar imbalance. The total strength of all peripheral classes is

$$m \sum_{i=1}^{n} r_i = m \sum_{i=1}^{n} (n+1+i) = \frac{3mn(n+1)}{2}.$$

Subtracting this from the total number of edges in (10) gives the total core strength

$$|E(\mathrm{CNC}_m(n))| - m \sum_{i=1}^{n} r_i = m(n+1). \tag{14}$$

□

For notational economy, define the normalized geometric-arithmetic factor

$$h(x,y) = \frac{2\sqrt{xy}}{x+y} \qquad (x, y > 0).$$

Thus $h(x,y)$ is exactly the local GA_e contribution of a quotient edge whose two edge-side strengths are x and y, before multiplication by the quotient edge strength.

For $m \ge 5$, $n \ge 1$, and $1 \le i \le n$, the quantities B_i and C_i in Lemma 4.1 are positive. Indeed, $B_i \ge B_1 = 2n+2 > 0$, and C_i is decreasing in i, so

$$C_i \ge C_n = \frac{m(n+1)(3n+2) - 9n^2 - 5n}{2} \ge 3n^2 + 10n + 5 > 0.$$

Thus every factor $h(B_i, C_i)$ used below is well-defined.

Theorem 4.2. *For integers $m \geq 5$ and $n \geq 1$,*

$$\mathrm{GA}_e(\mathrm{CNC}_m(n)) = m(n+1) + m\sum_{i=1}^{n}(n+1+i)h(B_i, C_i), \tag{15}$$

where B_i and C_i are given by (12) and (13).

Proof. Consider one peripheral class $E_{j,i}$. By Lemma 4.1, its quotient graph consists of a single quotient edge with strength $r_i = n+1+i$ and side strengths B_i and C_i. By the weighted definition (5), the contribution of this quotient edge is

$$r_i \frac{2\sqrt{B_iC_i}}{B_i + C_i} = r_i h(B_i, C_i) = (n+1+i)h(B_i, C_i).$$

For a fixed level i there are m symmetric choices of j, so the total peripheral contribution at level i is

$$m(n+1+i)h(B_i, C_i).$$

Summing over $i = 1, \ldots, n$ gives the peripheral part

$$m\sum_{i=1}^{n}(n+1+i)h(B_i, C_i).$$

For a core class, the two quotient side strengths are equal, say a and a. Hence its local factor is $h(a, a) = 1$, and its contribution is simply its edge strength. By (14), the sum of all core strengths is $m(n+1)$. Adding the core and peripheral contributions proves (15). □

4.1 Exact edge-Mostar expression

The same quotient data yield the exact edge-Mostar formula

$$\mathrm{Mo}_e(\mathrm{CNC}_m(n)) = m\sum_{i=1}^{n}(n+1+i)|B_i - C_i|. \tag{16}$$

Indeed, for a peripheral class $E_{j,i}$ the quotient edge has strength $r_i = n+1+i$ and side strengths B_i and C_i, so its edge-Mostar contribution is $r_i|B_i - C_i|$. There are m symmetric classes at level i. The core classes do not appear in (16) because they are edge-balanced.

For later use, define

$$\Delta_i = C_i - B_i = \frac{D_i}{2}, \tag{17}$$

where

$$D_i = m(n+1)(3n+2) - 6i^2 - (12n+8)i + 2n + 2. \tag{18}$$

The sequence D_i is strictly decreasing, since

$$D_{i+1} - D_i = -12i - 12n - 14 < 0.$$

Hence its sign can change at most once. To locate this possible sign change, regard D_i as the value at $x = i$ of the quadratic function

$$D(x) = m(n+1)(3n+2) - 6x^2 - (12n+8)x + 2n + 2.$$

The positive root of $D(x) = 0$ is

$$\rho_{m,n} = \frac{-(12n+8) + \sqrt{(12n+8)^2 + 24\{m(n+1)(3n+2) + 2n+2\}}}{12}. \tag{19}$$

Thus $D_i \geq 0$ precisely for the integer levels i not exceeding this root. We therefore set

$$s = \max\{0, \min\{n, \lfloor \rho_{m,n} \rfloor\}\}. \tag{20}$$

Then $D_i \geq 0$ for $1 \leq i \leq s$ and $D_i < 0$ for $s < i \leq n$. Consequently, an exact closed expression for the edge-Mostar index is

$$\mathrm{Mo}_e(\mathrm{CNC}_m(n)) = \frac{m}{2}\left(2\sum_{i=1}^{s}(n+1+i)D_i - \sum_{i=1}^{n}(n+1+i)D_i\right). \tag{21}$$

Using the standard sums $\sum_{i=1}^{n} i = n(n+1)/2$, $\sum_{i=1}^{n} i^2 = n(n+1)(2n+1)/6$, and $\sum_{i=1}^{n} i^3 = n^2(n+1)^2/4$, one obtains

$$\sum_{i=1}^{n}(n+1+i)D_i = \frac{n(n+1)}{6}\big(27mn^2 + 45mn + 18m - 81n^2 - 97n - 20\big). \tag{22}$$

Corollary 4.3. *If $D_n \geq 0$, equivalently if $B_i \leq C_i$ for all $1 \leq i \leq n$, then*

$$\mathrm{Mo}_e(\mathrm{CNC}_m(n)) = \frac{mn(n+1)}{12}\big(27mn^2 + 45mn + 18m - 81n^2 - 97n - 20\big). \tag{23}$$

In particular, (23) holds for every $n \geq 1$ when $m \geq 6$.

Proof. If $D_n \geq 0$, the monotonicity of D_i implies $D_i \geq 0$ for every i. Thus the absolute value in (16) may be removed, and (22) gives (23). For $m \geq 6$,

$$D_n = m(3n^2 + 5n + 2) - 18n^2 - 6n + 2 \geq 24n + 14 > 0.$$

□

Remark 4.4. *For $m = 5$, the condition $D_n \geq 0$ is equivalent to $-3n^2 + 19n + 12 \geq 0$, hence it holds for $1 \leq n \leq 6$. For pentagonal cones with $n \geq 7$, the exact absolute-value formula (16) or (21) should be used.*

4.2 Pentagonal, hexagonal, and heptagonal cones

The most common small-core subfamilies are $\mathrm{CNC}_5(n)$, $\mathrm{CNC}_6(n)$, and $\mathrm{CNC}_7(n)$. The edge-geometric-arithmetic formulas for these families are obtained by substituting $m = 5, 6, 7$ into (15). Thus

$$\mathrm{GA}_e(\mathrm{CNC}_m(n)) = m(n+1) + m\sum_{i=1}^{n}(n+1+i)\frac{2\sqrt{B_i C_i^{(m)}}}{B_i + C_i^{(m)}}, \qquad m \in \{5, 6, 7\}, \tag{24}$$

where

$$C_i^{(m)} = \frac{m(n+1)(3n+2) - 3i^2 - (6n+5)i}{2}.$$

For the edge-Mostar index, (16) gives the pentagonal formula

$$\mathrm{Mo}_e(\mathrm{CNC}_5(n)) = 5\sum_{i=1}^{n}(n+1+i)|B_i - C_i^{(5)}|. \tag{25}$$

For $1 \leq n \leq 6$, this reduces to

$$\mathrm{Mo}_e(\mathrm{CNC}_5(n)) = \frac{5n(n+1)(27n^2 + 64n + 35)}{6}. \tag{26}$$

For all $n \geq 1$, the hexagonal and heptagonal cases are sign-stable and therefore

$$\mathrm{Mo}_e(\mathrm{CNC}_6(n)) = \frac{n(n+1)(81n^2 + 173n + 88)}{2}, \tag{27}$$

and

$$\mathrm{Mo}_e(\mathrm{CNC}_7(n)) = \frac{7n(n+1)(54n^2 + 109n + 53)}{6}. \tag{28}$$

5 Numerical comparison

Table 1 reports exact edge-Mostar values and decimal approximations of GA_e for $n = 1, \ldots, 10$. The pentagonal values for $n \geq 7$ use the absolute-value expression (25).

Table 1: Comparison of GA_e and Mo_e for $\mathrm{CNC}_5(n)$, $\mathrm{CNC}_6(n)$, and $\mathrm{CNC}_7(n)$.

n	$\mathrm{GA}_e(\mathrm{CNC}_5)$	$\mathrm{Mo}_e(\mathrm{CNC}_5)$	$\mathrm{GA}_e(\mathrm{CNC}_6)$	$\mathrm{Mo}_e(\mathrm{CNC}_6)$	$\mathrm{GA}_e(\mathrm{CNC}_7)$	$\mathrm{Mo}_e(\mathrm{CNC}_7)$
1	21.571	210	24.789	342	27.890	504
2	50.833	1355	58.210	2274	65.111	3409
3	92.431	4700	105.826	8016	118.160	12124
4	146.342	12050	167.599	20760	186.990	31570
5	212.560	25750	243.519	44670	271.586	68180
6	291.082	48685	333.582	84882	371.943	129899
7	381.906	84430	437.785	147504	488.060	226184
8	485.033	138880	556.128	239616	619.934	368004
9	600.462	215550	688.610	369270	767.564	567840
10	728.192	319665	835.230	545490	930.950	839685

The two descriptors behave differently because their local summands have different scales. The edge-geometric-arithmetic contribution of a bond is at most 1, so $\mathrm{GA}_e(\mathrm{CNC}_m(n)) \leq |E(\mathrm{CNC}_m(n))| = O(n^2)$ for fixed m. Formula (15) indeed shows quadratic growth. By contrast, in the sign-stable range $m \geq 6$, Equation (23) gives

$$\mathrm{Mo}_e(\mathrm{CNC}_m(n)) = \frac{9}{4}m(m-3)n^4 + O(n^3), \qquad n \to \infty. \tag{29}$$

Thus the edge-Mostar index is much more sensitive to the increase in the number of layers. Chemically, this means that Mo_e emphasizes the accumulated edge-distance nonbalancedness of larger cones, whereas GA_e measures the average local balance of bond environments after normalization.

It is also useful to interpret (3). A peripheral class for which B_i and C_i are close contributes nearly r_i to GA_e but little to Mo_e. A class with a large difference $|B_i - C_i|$ contributes strongly to Mo_e but weakly to GA_e. Hence the two indices should not be viewed as substitutes; rather, they provide complementary information about edge-distance balance.

6 Conclusion

We introduced the edge-geometric-arithmetic index GA_e, an edge-distance analogue of the second geometric-arithmetic index, and established its quotient formula for strength-weighted quotient graphs. Applying this formula to carbon nanocones produced the exact expression

$$\mathrm{GA}_e(\mathrm{CNC}_m(n)) = m(n+1) + m\sum_{i=1}^{n}(n+1+i)\frac{2\sqrt{B_iC_i}}{B_i + C_i}.$$

The corresponding edge-Mostar index is exactly

$$\mathrm{Mo}_e(\mathrm{CNC}_m(n)) = m\sum_{i=1}^{n}(n+1+i)|B_i - C_i|.$$

When the side differences do not change sign, this expression reduces to a polynomial; this includes all hexagonal and heptagonal cones and, more generally, all $m \geq 6$. For pentagonal cones with sufficiently many layers, the absolute-value expression is the appropriate all-parameter formula.

The comparison shows a clear conceptual distinction. The edge-Mostar index is an unnormalized measure of edge-distance nonbalancedness, while GA_e is a bounded local balance score summed over bonds. Future work may extend this approach to total-geometric-arithmetic indices, coronoid systems, nanotubes, and benzenoid networks, where quotient methods can again convert large molecular graphs into tractable weighted quotient graphs.

Declarations

Data availability. No experimental data are used. All numerical values in Table 1 are obtained directly from the formulas in this article.

Conflict of interest. The author(s) declare no conflict of interest.

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